of individuals who report hearing problems seek professional help through the use of hearing aids (Abrams & Kihm, 2015). Reasons for this are many and can include factors such as denial (e.g., "My hearing loss is not that bad," "I can get along okay without them"), consumer issues (e.g., "I heard hearing aids do not work"), stigma (e.g., "Hearing aids will make me look old"), cost, cosmetic issues, other health issues, personality, and attitudes of others. The audiologist should provide the client with recommendations that will improve their quality of life, but the client will make the final decisions.

Hearing Aid Fitting Protocol

The American Academy of Audiology Guidelines for the Audiologic Management of Adult Hearing Impairment (Valente et al., 2006), which was listed as "in revision" when this chapter was being revised, as well as the College of Speech and Hearing Professionals of British Columbia (2016) Clinical Practice Guideline: Audiologic Management of Adult Hearing Impairment, list four critical areas to be addressed for every hearing aid fitting: selection, quality control, the fitting, and verification.

Selection

The recommendation for hearing aid style must be based on the gain and output needs for the hearing loss, audiogram configuration, the manipulation skills of the patient, the need for various features such as directional microphones, comfort, and patient preferences (cosmetic concerns, prior experience, etc.).

In general, the more severe the hearing loss, the more gain is needed. BTE instruments can provide greater gain without acoustic feedback (whistling) than can smaller ITE or ITC devices, whereas some ITE and ITC devices may be perceived as less visible. Open-fit BTE devices allow for low-frequency information to be delivered through the normal pathway while providing amplification to mid and high frequencies through the hearing aid for patients with a moderate high-frequency loss and normal hearing in the low frequencies. These devices have been well received due to their small size and comfort. In general, two hearing aids are better than one. Bilateral hearing aids can provide the listener better speech-understanding ability in noisy settings. Most patients are able to localize where sounds are coming from better with two hearing aids and will feel more balanced when fit bilaterally. A small number of individuals may not receive these benefits from two hearing aids and, in fact, may function more poorly with the second hearing aid. Persons with certain auditory processing difficulties, such as binaural inference, or with asymmetric hearing losses also may do better with one hearing aid. Sometimes, financial or cosmetic concerns also prevent patients from opting for two versus one hearing aid. In most cases, however, it is best to try two hearing aids first because the vast majority of patients are successful bilateral users.

Selection of the type of digital signal processing (DSP) is also important and should be based on the audiometric needs of the patient. The number of frequency bands or channels should be determined by the frequency shaping that is needed to provide the best audibility of the speech signal. Flatter-shaped audiograms with a similar loss across the speech frequencies may require fewer channels than those with sloping or unusual audiometric configurations.

Compression characteristics need to be considered based on the patient's dynamic ranges, which may vary by frequency. Persons with sensorineural hearing loss typically have recruitment. This means that they have smaller dynamic ranges and therefore need more gain or amplification for softer sounds and less gain or amplification for louder sounds. Compression

Dynamic range is the difference between the hearing aid user's pure-tone or speech thresholds and their uncomfortable listening level. Many individuals with hearing loss have a reduced dynamic range as compared to individuals with normal hearing.

TABLE 3–1. Overview of U.S. Food and Drug Administration Changes to Cochlear Implant Candidacy Guidelines Through the Years in Age at Implantation, Onset and Degree of Hearing Loss, and Aided Open-Set Speech Perception Scores (for Adults Only)

YEAR	1985	1990	1998	2000	2014	2019	2020	2022
AGE	**adults** (18 yrs. and older)	**adults & children** (2 yrs. and older)	**adults & children** (18 mos. and older)	**adults & children** (12 mos. and older)	. . .	. . .	**adults & children** (12 mos. and older) Advanced Bionics/ MED-EL ---- (9 mos. and older) Cochlear Corporation	. . .
ONSET of HEARING LOSS	postlingual	postlingual adults pre- & postlingual children	. . .	. . .	. . .	. . .	. . .	. . .

YEAR	1985	1990	1998	2000	2014	2019	2020	2022
DEGREE of SNHL	profound	**adults and children:** profound	**adults:** severe-profound --- **children:** profound	**adults and children older than 24 mos.:** bilateral severe-profound --- **children between 12–24 mos. old:** bilateral profound	**<u>electroacoustic (EAS, hybrid, hearing preservation, etc.)</u>** **adults only:** typical to moderate low-frequency hearing loss (thresholds better than 60 dB HL up to and including 500 Hz). Severe to profound, mid- to high-frequency hearing loss in ear to be implanted. --- **children 24 mos. and older:** bilateral, severe-profound --- **children aged 9–24 mos.:** bilateral, profound	**<u>single-sided (SSD)</u>** **adults and children older than 5 yrs.:** ear to be implanted: severe to profound hearing loss (PTA at 500, 1000, 2000, and 4000 Hz of >80 dB HL) contralateral ear: typical or near-typical hearing (PTA at 500, 1000, 2000, and 4000 Hz ≤30 dB HL) **<u>asymmetrical (AHL)</u>** **adults and children older than 5 yrs.:** ear to be implanted: same as SSD contralateral ear: mild to moderately severe hearing loss with a difference of at least 15 dB ---- **children >24 mos:** bilateral severe-profound ---- **children 9–24 mos.:** bilateral profound	. . .	. . .
OPEN-SET SPEECH SCORES (ADULTS ONLY)	0%	0%	40% or less (CID sentences)	<50% (HINT) in ear to be implanted with <60% in the best-aided condition	60% CNC word recognition score in ear to be implanted Medicare: 40% CNC word recognition score or less with connected speech; best aided condition	5% CNC word recognition score in ear to be implanted 90% CNC word recognition score in contralateral ear Medicare: 40% or less connected speech; best-aided condition		Medicare: 60% or less with connected speech; best-aided condition

Note: SNHL = sensorineural hearing loss; mos. = months; yrs. = years; dB = decibels; HL = hearing level; PTA = pure-tone average; Hz = hertz; CID = Central Institute for the Deaf; HINT = Hearing in Noise Test; CNC = consonant-nucleus-consonant; . . . = no change.

The Auditory Training Process

Auditory training is a major component of the audiologic rehabilitation process. Thus, its potential in assisting those with hearing loss has been expressed in major textbooks within the field of audiology both in the past (Davis & Hardick, 1981; Oyer, 1966) and more recently (Boothroyd, 2007; Cole & Flexer, 2015; Montano & Spitzer, 2013; Tye-Murray, 2024). See Chapter 1 (Table 1–4) for a model to understand AR and how auditory training fits into the overall assessment and management of AR. The intent of the next major section of this chapter is to familiarize the reader with both the traditional and the current forms of auditory training and how they fit into the entire audiologic rehabilitation process.

Definition and Application of Auditory Training

Numerous attempts have been made to define auditory training in the past. Definitions of auditory training can vary considerably according to the orientation of the definer and special considerations associated with hearing loss, such as its degree and time of onset. With the decreased age of identification of hearing loss and improvements in hearing technology, auditory training today includes a variety of aspects, such as caregiver training and the ability to make changes within the hearing technology, that haven't been included in the past. Historically, the definition of auditory training is attributed to Carhart (1960), who considered auditory training a process of teaching the child or adult with hearing loss to take full advantage of available auditory clues. As a result, Carhart recommended an emphasis in therapy on developing an awareness of sound, gross discrimination of nonverbal stimuli, and gross and fine discrimination of speech.

Later, in discussing the use of auditory training with children, Erber (1982) described it as "the creation of special communication conditions in which teachers and audiologists help hearing-impaired children acquire many of the auditory perception abilities that normally hearing children acquire naturally without their intervention" (p. 1). Erber stated further,

> Our intent is to help the hearing-impaired child apply their impaired auditory sense to the fullest capacity in language communication, regardless of the degree of damage to the auditory system. Usually progress is achieved through careful application of amplification devices and through special teaching techniques. (p. 29)

When considering auditory training for adults, two general objectives are usually relevant: (a) learning to maximize the use of auditory and other related cues available for the perception of speech and (b) adjustment and orientation to facilitate the optimum use of hearing aids and cochlear implants.

The ultimate aim of auditory training is to optimize communication potential through developing the auditory sensory channel, with the use of appropriately fit hearing technology, to its fullest. For adults, auditory training is often designed to improve one's listening skills, resulting in improved speech perception abilities in a variety of communicative contexts. It is important to point out that improved speech perception skills can often lead to more proficient speech and language skills, educational and vocational advancement, and successful psychosocial adjustment. For children, auditory training is part of a comprehensive communication development focused not only on receptive communication (or speech perception) but also on integration of complex expressive language across domains (as addressed in more detail in Chapter 9).

The acoustic hearing abilities that a person with hearing loss has left are often referred to as "residual hearing."

Early Efforts in Auditory Training

The earliest efforts in auditory training date back at least to the 19th century. Individuals in Europe used auditory training with those having hearing loss throughout the 1800s, with some success noted. Impressed with their accomplishments, Goldstein (1939) introduced a similar approach to auditory training in the United States in the late 1890s and early 1900s. Known as the Acoustic Method, this approach centered around systematic stimulation with individual speech sounds, syllables, words, and sentences to improve speech perception and to aid deaf persons in their own speech production. The Acoustic Method was utilized in a number of facilities throughout the country, including the Central Institute for the Deaf in St. Louis, Missouri, which Goldstein founded. Goldstein exerted a significant influence on the thinking of many professionals over the years regarding the potential of auditory training with persons with hearing loss.

Until World War II, the primary focus of auditory training was its use with severely/profoundly d/Deaf children to facilitate speech and language acquisition and increase their educational potential. However, the activities that occurred at Veterans Administration audiology centers during World War II demonstrated on a large scale that adults with mild to severe hearing losses could also benefit from auditory training. Led by Raymond Carhart, personnel in these centers developed and applied auditory training exercises with large numbers of adults, most with noise-induced hearing loss. Carhart (1960) later authored a book chapter that influenced the thinking of audiologists regarding auditory training with children and adults for years to come.

Carhart

Raymond Carhart made many contributions to the profession of audiology.

Carhart was one of the pioneers who considered auditory training both for children and adults, based on the idea that auditory cues could be used to improve spoken communication.

Childhood Procedures

Carhart outlined four major steps or objectives involved in auditory training for children with prelingual deafness:

1. Development of awareness of sound
2. Development of gross discriminations
3. Development of broad discriminations among simple speech patterns
4. Development of finer discriminations for speech

Development of an awareness of auditory stimuli and the significance of sound involves having the child acknowledge the presence and absence of sound and the importance of sound in their world. The development of gross discrimination initially involves demonstrating with various noisemakers that sounds differ. Once the child can successfully discriminate grossly different sounds, they are exposed to finer types of discrimination tasks that include variation in the frequency, intensity, and durational properties of sound. When the child can recognize the presence of sound and perceive gross differences with nonverbal stimuli, Carhart's approach calls for the introduction of activities directed toward learning gross discrimination for speech signals. The final phase consists of training the child to make fine discriminations of speech stimuli in connected discourse and integrating an increased vocabulary to enable them to follow connected speech in a more rapid and accurate fashion. Carhart also felt that the child should be encouraged to use vision in most auditory training activities. Today, early

intervention and advanced hearing technology builds upon these foundational aspects, but infants and young children are better able to move through detection of sound at a more rapid pace.

Adult Procedures

Because adults who acquire a hearing loss later in life retain a portion of their original auditory skills, Carhart recommended that auditory training with adults focus on reeducating a skill diminished as a consequence of the hearing loss. Initially, Carhart felt that it was important to establish "an attitude of critical listening" in the individual. This involves being attentive to the subtle differences among sounds and can involve a considerable amount of analytic drill work on the perception of phonemes that are difficult for the adult with hearing loss to perceive. Lists of matched syllables and words that contain troublesome phonemes, such as "she"/"fee," "so"/"though," "met"/"let," or "mash"/"math," are read to the individual, who repeats them back. Such training should also include phrases and sentences, with the goal of developing as rapid and precise a recognition of the phonetic elements as is possible within the limitations imposed by the person's hearing loss. Carhart also encouraged speechreading combined with a person's hearing during a portion of the auditory training sessions.

Analytic auditory training employs a bottom-up approach that focuses mainly on enhancing perception of small segments of speech (phonemes, syllables, words).

Because we often communicate under less-than-ideal listening circumstances, Carhart advocated that auditory training sessions for adults be conducted in three commonly encountered situations: (a) relatively intense background noise, (b) the presence of a competing speech signal, and (c) listening on the telephone. This emphasis on practice in speech perception under listening conditions with decreasing amounts of redundancy has been emphasized, recently, as noted by Sweetow and Henderson-Sabes (2010).

According to Carhart, the use of hearing technology is vital in auditory training, and he recommended that technology be utilized as early as possible in the auditory training program. These recommendations are consistent with Carhart's belief that systematic exposure to sound during auditory training is an ideal means of allowing a person to adequately adjust to hearing aids and assist in using them as optimally as possible.

Current Approaches to Auditory Training

The basic intent of more recent methods of auditory training remains the same: to maximize communication potential by developing to its fullest the auditory channel of the person with hearing loss. This next section discusses how this form of audiologic rehabilitation is currently being applied with those experiencing hearing loss.

Candidacy for Auditory Training

Auditory training is beneficial for infants and young children with cochlear implants and/or hearing aids to help the child understand and make use of sound as part of spoken language development and communication. Because children are identified at earlier ages, auditory training has evolved to help to integrate listening skills into daily communication and early language learning and often includes caregivers as an integral part of supporting auditory- and language-rich communicative environments.

Adults with newly fit hearing technology can benefit from auditory training. There is strong evidence that a structured program of listening training enhances the benefits derived from a cochlear implant (Bernstein et al., 2021; Moberly et al., 2020; Völter et al., 2020).

Assessment of Auditory Skills

An integral part of a comprehensive auditory training program is the assessment of individual auditory skills. Before, during, and at the conclusion of auditory training, the clinician should attempt to evaluate the auditory abilities of the person with hearing loss. Information of this nature is important for several reasons, including the following:

- Determining how auditory training can support current functional listening skills
- Presurgery or prior to initiating aural rehabilitation baseline comparisons to posttherapy performance to assess how much improvement in auditory performance, particularly speech perception, has occurred
- Identifying specific areas of auditory skill development and perception of spoken language to focus auditory training

There are many factors and variables that determine the auditory testing that takes place, such as the age of the client, the language skills, and the type and degree of the hearing loss. The clinician must exercise care in selecting appropriate test materials for the individual patient, particularly with regard to the language levels required for a given test. When testing speech perception performance, a number of variables need to be kept in mind that can influence the degree of difficulty of the perceptual task. These include the following:

- The nature of the perceptual task (detection, discrimination, identification, comprehension)
- The use of either an open-set or a closed-set (e.g., multiple-choice) response format
- The degree of context present (e.g., monosyllabic words alone; without sentences)
- The level of sophistication and the age level of the vocabulary used within the test (instructions and test items)
- The signal-to-noise ratio present (figure–ground)
- Isolation of the hearing technology to assess acuity specifically with use of that device

Figure–ground refers to the ability to perceive a target signal that is presented simultaneously with other competing signals.

Manipulation of any one of these variables can make the test either easier or harder for an individual, and the clinician should be mindful of all this when selecting and administering tests. It is important to obtain a full picture of current auditory abilities. A variety of tests should be available for assessment purposes so that the particular needs of each individual can be determined adequately.

Evaluating Children

Informal testing, observation, and caregiver reports are an integral aspect of assessing the auditory skills of young children. For the infant, the initial goal of assessment for auditory training will focus on detection of sounds across all frequencies. It is particularly important to assess how well the hearing technology is fit for the child's particular hearing loss and to make programming changes as needed. Once this information is known, a specific program for developing auditory skills, such as that described in Chapter 9, can be implemented in conjunction with therapy related to development of speech and language. It is important that audiologists, parents/caregivers, and early interventionists collaborate to ensure the child's hearing technology is well fit and provides the child with the acoustic access needed to develop spoken communication. Parent-completed questionnaires, such as the LittlEARS (Tsiakpini et al., 2004), the Parent's Evaluation of Aural/Oral Performance of Children



Informal assessment of auditory skills.

(PEACH) (Ching & Hill, 2005), and the Infant-Toddler Meaningful Auditory Integration Scale (ITMAIS) (Zimmerman-Phillips et al., 1997), ask caregivers to rate and/or report on observed behaviors. The behaviors are interpreted as responses to auditory stimuli. These questionnaires help provide audiologists with a framework of understanding how the hearing technology is working and whether any programming changes are needed.

For older children, more formal, in-depth assessment of overall speech perception abilities is possible. The Pediatric Minimum Speech Test Battery (PMSTB) measures are organized in a hierarchical structure based on difficulty (Uhler et al., 2017). Specifically, materials have been developed that require the child to respond in a prescribed manner to individual words or phonemes presented at a comfortable listening level (Figure 4–6).

Some of the commonly used closed-set measures include formal tests that have been designed for assessing speech perception in children with hearing loss, such as the following:

Tests use either a closed-set or an open-set format. A closed-set format for a test of speech perception involves presenting a test item (e.g., a word) and having the listener choose the correct response from a limited set of options (multiple choice). In an open-set format, the listener can respond with any word they feel is correct.

- Word Intelligibility by Picture Identification (WIPI) (Ross & Lerman, 1970). The authors of the WIPI modified an existing test for children (Myatt & Landes, 1963) to include only vocabulary appropriate for children with hearing loss. The WIPI includes four lists, each of which contains 25 monosyllabic words. The child provides a picture-pointing response in a closed-set format. According to the authors, the test is suitable for use with children with hearing loss and limited receptive and expressive language abilities.
- Northwestern University Children's Perception of Speech (NU-CHIPS) (Katz & Elliott, 1978). This test consists of 50 monosyllabic nouns that have been scrambled to form four individual lists. The NU-CHIPS uses a response format that requires that the child point to the one picture from several options that best represents the test items. The NU-CHIPS includes basic vocabulary and allows for a nonverbal response format.

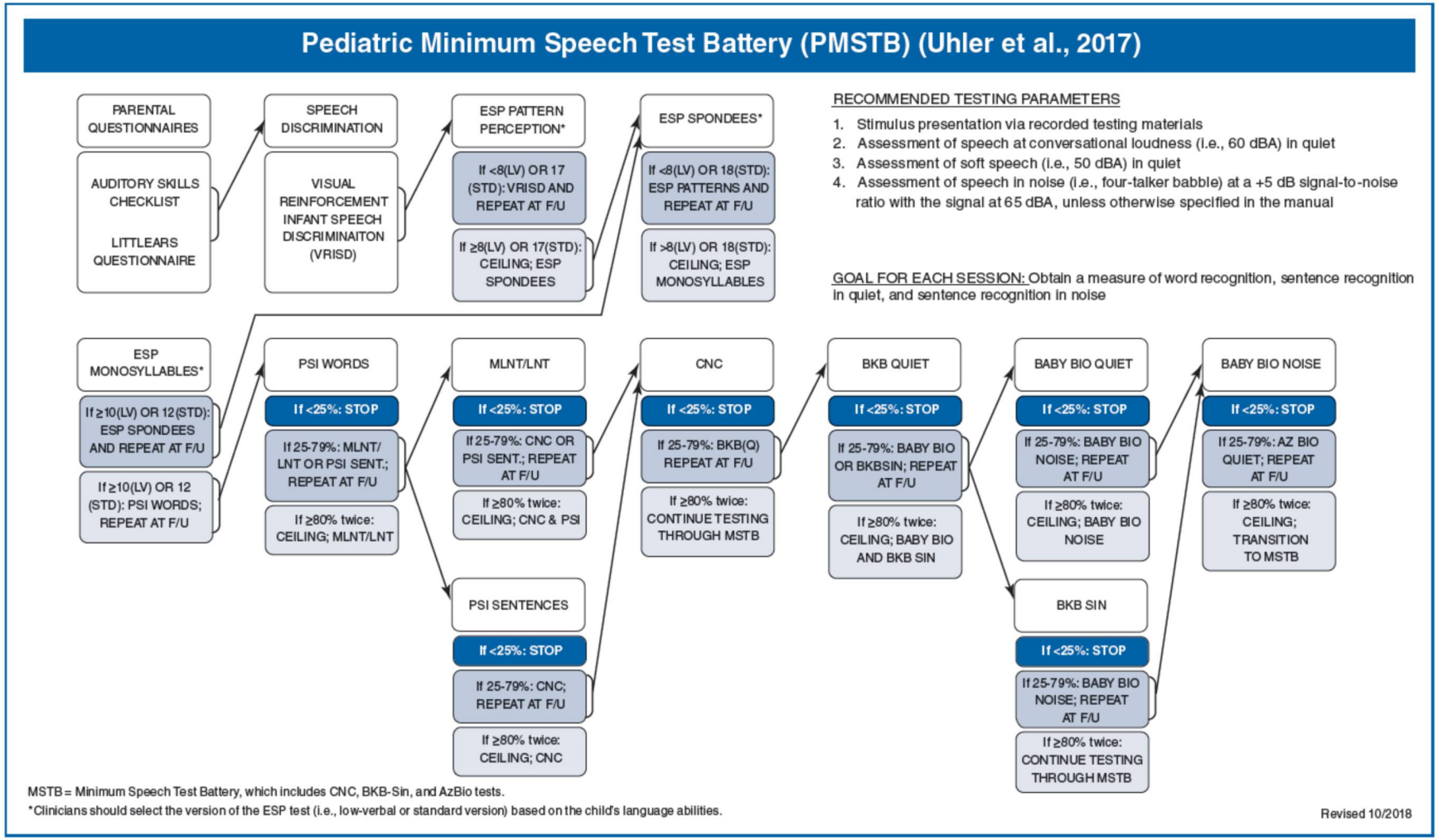


FIGURE 4-6. An example of the hierarchical structure of the Pediatric Minimum Speech Test Battery (PMSTB). Source: Uhler et al. (2017).

- Six Sound Test (Ling, 1976, 1989). Six isolated phonemes (/m/, /a/, /u/, /i/, /s/, and /ʃ/) are spoken to the child at a normal conversational level. In 2021, the 10 Sound Test expanded the phonemes included with the addition of /z/, /h/, /n/, and /ʤ/ (Madell & Hewitt, 2021).

Additional test batteries are designed to assess varied aspects of auditory skill development in children. Examples include the following:

- Early Speech Perception Test (ESP) (Moog & Geers, 1990). ESP consists of a series of subtests that assess pattern and word perception using a closed-set format. The results allow children to be placed into four speech perception categories: (a) no pattern perception, (b) pattern perception, (c) some word identification, and (d) consistent word identification. A low-verbal version was developed for use with children with restricted vocabulary.
- Glendonald Auditory Screen Procedure (GASP), developed at the Glendonald Auditory School for the Deaf in Australia (Erber, 1982). GASP is based on a model of auditory perception described in the next section of this chapter (Figure 4–7). The basic test battery associated with GASP consists of three subtests of speech perception: (a) phoneme detection, (b) word identification, and (c) sentence comprehension. The GASP phoneme detection subtest is similar in format to the Six Sound Test developed by Ling (1976, 1989). According to Erber, the results from GASP can aid in planning auditory training because the child's performance on the subtests is predictive of other, related auditory tasks.
- Developmental Approach to Successful Listening Test (DASL II) (Stout & Windle, 1994). This comprehensive test of auditory skills evaluates numerous aspects of sound awareness, phonetic listening, and auditory comprehension. Children from 3 years of age can be evaluated with the DASL Test, and some normative information is available for children with varying degrees of hearing loss.

All these tests attempt to take into account potential limitations of a child's receptive vocabulary level and ability to respond orally. However, the variability observed in the receptive and expressive communication skills of these children makes it unwise to draw any firm generalization about the specific age range of children for whom any of these tests are suited. Vocabulary development and expressive language skills, rather than chronological age, is a key consideration in selecting the appropriate test to use.

Evaluating Adults

A number of formal tests of speech perception are available for use with adults. The minimum speech test battery (MSTB) for postlingually deafened adult cochlear implant patients includes the Hearing in Noise Test (HINT) for sentence level and the Consonant/Nucleus/Consonant (CNC) test for word level (Luxford et al., 2001).

Word recognition testing typically involves presenting a 25- or 50-word list of monosyllabic words at a comfortable intensity level for the listener. A percent correct score is calculated.

Historically, other tests allow for more in-depth assessment of the perception of consonants, which can be especially difficult for persons with hearing loss to perceive accurately. For example, Owens and Schubert (1977) produced a 100-item, multiple-choice consonant perception test called the California Consonant Test (CCT). Thirty-six of the test items assess consonant perception in the initial word position, while 64 items test perception in the final position. Each of the 100 items is presented to the listener in a closed-set, multiple-choice format, as shown in Figure 4–8. Research by Schwartz and Surr (1979) demonstrated that, compared to the Northwestern University Auditory Test No. 6, the CCT is more sensitive to

Speech Stimulus

Response Task	Speech Elements	Syllables	Words	Phrases	Sentences	Connected Discourse
Detection	1					
Discrimination						
Identification			2			
Comprehension					3	

FIGURE 4-7. An auditory stimulus–response matrix showing the three GASP subtests: Phoneme Detection (1), Word Identification (2), and Sentence (Question) Comprehension (3). Reprinted with permission from the Association for the Deaf.

ROBE _____	MAP _____	BAIL _____
RODE _____	MATCH _____	JAIL _____
ROSE _____	MATH _____	DALE _____
ROVE _____	MAT _____	GALE _____
LASS _____	DIES _____	LEAF _____
LAUGH _____	DIED _____	LEASE _____
LATCH _____	DIVE _____	LEACH _____
LASH _____	DINE _____	LEASH _____
FIN _____	PEAK _____	RAISE _____
PIN _____	PEACH _____	RAID _____
KIN _____	PEAT _____	RAGE _____
TIN _____	PEEP _____	RAVE _____

FIGURE 4-8. Examples of multiple-choice test items for the California Consonant Test. Source: Owens and Schubert (1977).

the speech recognition difficulties experienced by individuals with high-frequency hearing loss. Consequently, the CCT is frequently used to assess the speech recognition abilities of adult patients.

Tests that employ sentence-type stimuli also can be informative. Kalikow et al. (1977) developed a test called Speech Perception in Noise (SPIN).

This test is unique in that it attempts to assess a listener's utilization of both linguistic and situational cues in the perception of speech. Sentence material is presented against a background of speech babble, with the listener's task being to identify the final word in the sentence. Ten 50-item forms of SPIN have been generated, with each version containing sentences with either high or low predictability relative to the final word in each sentence. Examples of each are shown in Table 4–6. Bilger et al. (1979) revised the forms to make them more equivalent to each other. The SPIN test can provide important information concerning how effectively a given listener uses contextual information in the perception of speech, in addition to providing insight into how the listener perceives the acoustical properties of speech.

Speech babble is a recording of several people talking at once.

The Central Institute for the Deaf (CID) Everyday Speech Sentences (Davis & Silverman, 1978) has been used extensively for several decades to evaluate a listener's ability to perceive connected discourse. They consist of 10 sets of 10 sentences, as shown in Table 4–7. The

TABLE 4–6. Examples of Low- and High-Predictability Sentences From the SPIN Test

Sentence	Level of Predictability
The honeybees swarmed around the ***hive***.	High
The girl knows about the ***swamp***.	Low
The cushion was filled with ***foam***.	High
He had considered the ***robe***.	Low

Source: Kalikow et al. (1977).

TABLE 4–7. Examples of a 10-Sentence set of CID Everyday Speech Sentences

1. *It's time* to *go.*
2. *If* you *don't want these old magazines, throw them out.*
3. *Do* you *want to wash up?*
4. It's a *real dark night so watch your driving.*
5. *I'll carry* the *package* for *you.*
6. Did *you forget* to *shut off* the *water?*
7. *Fishing* in a *mountain stream* is my *idea* of a *good time.*
8. *Fathers spend* more *time* with their *children than* they *used to.*
9. *Be careful not to break your glasses!*
10. *I'm sorry.*

Source: Davis and Silverman (1978).

sentences vary in length and form and possess several characteristics associated with typical conversation.

Results of these tests, as well as others, should provide the clinician with specific information concerning a client's consonant perception in a word and/or sentence context as well as the ability to comprehend speech in sentence form.

Additional information about speech perception can be gained by introducing competing noise to the test situation and varying the degree of redundancy in the test material. Commonly used tests of this type include the Hearing in Noise Test (HINT) (Nilsson et al., 1994) and Quick Speech in Noise (QuickSIN) (Etymotic Research, 2001). Also, addition of visual cues via speechreading during test administration in a bisensory perceptual condition can provide useful information regarding a person's overall integrative skills (see Chapters 5 and 10).

Methods of Auditory Training

The more current approaches to auditory training vary considerably. According to Blamey and Alcantara (1994), it is possible to categorize them into one of four general categories, based on the fundamental strategy stressed in therapy:

- Analytic: Attempts to break speech into smaller components (phoneme, syllable) and incorporate these separately into auditory training exercises. Examples include exercises that emphasize same–different discrimination of vowel or consonant phonemes in syllables (e.g., /bi-ba/) or words (e.g., /kip-kit/) or that require the listener to identify a word within a closed-set response format (e.g., "run"/"money"/"bat"). This is commonly known as a bottom-up approach to auditory skill development.
- Synthetic: Emphasizes a more global approach to speech perception, stressing the use of clues derived from the syntax and context of a spoken message to derive understanding. Training synthetically involves the use of meaningful stimuli (words, phrases, sentences). This might involve practicing sentence perception based on prior information about context (e.g., having lunch, a classroom discussion on government) or having the clinician name a topic and present related words or phrases that the individual must repeat back. This is commonly known as a top-down approach to auditory skill development.
- Pragmatic: Involves training the listener to control communication variables, such as the level of speech, the signal-to-noise ratio, and the context or complexity of the message, to obtain the necessary information via audition for understanding to occur. For example, the person with hearing loss practices how to effectively use conversation repair strategies, such as asking questions or requesting that a statement be repeated or clarified, to comprehend a paragraph read by the clinician. Here, the patient is given a series of questions related to a specific topic, such as those listed in Table 4–8. The patient asks the clinician each question. The clinician answers each question, and the patient then must correctly repeat the answer given before moving on to the next question. This conversationlike therapy strategy can be done in an auditory-only or auditory–visual perceptual mode.
- Eclectic: Includes training that combines most or all of the strategies previously described.

Signal-to-noise ratio refers to comparing the intensity of the signal you wish to hear with all the other auditory signals present in that listening situation.

TABLE 4–8. Topics and Questions From QUEST?AR QUESTions for Aural Rehabilitation

Where did you go?	museum, restaurant, post office, shopping, camping, doctor, zoo, beach, airport, swimming, mountains, picnic, music lesson, Mars, supermarket, and so forth
Questions:	1. Why did you go there? 2. When did you go? 3. How many people went with you? 4. Who were they? (names) 5. What did you take with you? 6. Where is (the place where you went)? 7. How did you get there? 8. What did you see on the way? 9. What time did you get there? 10. What did you do first? 11. What did you see? 12. How many? What color? etc. 13. What happened at (the place where you went)? 14. What else did you do? 15. What were other people doing at (the place where you went)? 16. What was the most interesting thing that you saw? 17. What was the most interesting thing that you did? 18. What did you buy? 19. What kind? What flavor? What color? etc. 20. How much did it cost? 21. Did anything unusual happen? What? 22. How long did you stay? 23. What did you do just before you came home? 24. When did you leave? 25. How did you get home? 26. What happened on the way home? 27. What time did you get home? 28. How did you feel then? 29. When are you going back? 30. Do you think that I should go sometime? Why?

Source: Erber (1996).

While all the auditory training programs to be described have analytic, synthetic, or pragmatic tendencies, most would best be described as eclectic since more than one general strategy for the training of listening skills is typically used with a given child or adult.

Erber

Erber (1982) described a flexible and widely used approach to auditory training designed for use primarily with children. This adaptive method is based on a careful analysis of a child's auditory perceptual abilities using the GASP assessment battery (described briefly in the earlier portion of this chapter devoted to assessment). Recall that GASP's approach to evaluating a child's auditory perceptual skills considers two major factors: (a) the complexity of the speech stimuli to be perceived (ranging from individual speech elements to connected discourse) and (b) the form of the response required from the child (detection, discrimination, identification, or comprehension). Several levels of stimuli and responses are involved, as shown in Figure 4–6. The GASP test battery initially evaluates only the three stimulus–response combinations indicated in the figure. However, Erber encourages the use of other available test materials to evaluate other stimulus–response combinations from the matrix in Figure 4–6, when appropriate.

Once the child's auditory capabilities are determined, an auditory training program is outlined using the same stimulus–response model as discussed for the GASP assessment when establishing goals and beginning points for therapy. Those stimulus–response combinations found not to be processed well during the GASP assessment phase logically become the same combinations targeted in auditory training activities that follow. Erber's approach is flexible and highly adaptable to children with a wide variety of auditory abilities since the stimulus and response combinations range from the simplest (phoneme detection) to the most complex (sentence comprehension) perceptual tasks.

Erber also described three general styles that the clinician may use during auditory training, depending on the communication setting. These styles differ in specificity, rigidity, and direction and are described in Table 4–9. Adaptive procedures, where the child's responses to speech stimuli are used to determine the next activity, can be employed with any of these styles. In attempting to develop a child's auditory abilities, Erber (1982) stated:

> Auditory training need not follow a developmental plan where, for instance, you practice phoneme detection first and attempt comprehension of connected discourse last. Instead, you might use the "conversational approach" during all daily conversation, and apply the "moderately structured approach" as a follow-up to each class activity. During each activity, you will note consistent errors. Later, you might provide brief periods of specific practice with difficult material. In this way, you can incorporate auditory training into conversation and instruction, rather than treat listening as a skill to be developed independently of communication. (p. 105)

Erber's emphasis on integrating the development of auditory skills into all activities with children with hearing loss is shared by many, including Sanders (1993) and Ling and Ling (1978), who recommend that auditory training "be viewed as a supplement to auditory experience and as an integral part of language and speech training" (p. 113). Thus, therapy directed toward the development of auditory and language skills can and should be done in an integrated, mostly seamless manner. As mentioned earlier, Erber's levels of perception model (detection, discrimination, identification, and comprehension) is widely used with both children and adults in rehabilitation therapy involving the development and improvement of auditory and visual perceptual skills for speech perception.

TABLE 4–9. Three General Auditory Training Approaches

Natural conversational approach	1. The teacher eliminates visible cues and speaks to the child in as natural a way as possible, while considering the general situational context and ongoing classroom activity. 2. The auditory speech perception tasks may be chosen from any cell in the stimulus–response matrix; for example, sentence comprehension. 3. The teacher adapts to the child's responses by presenting remedial auditory tasks in a systematic manner (modifies stimulus and/or response), derived from any cell in the matrix.
Moderately structured approach	1. The teacher applies a closed-set auditory identification task, but follows this approach activity with some basic speech development procedures and a related comprehension task. Thus, the method retains a degree of flexibility. 2. The teacher selects the nature and content of words and sentences on the basis of recent class activities. 3. A few neighboring cells in the stimulus–response matrix are involved (for example, word and sentence identification and sentence comprehension).
Practice on specific tasks	1. The teacher selects the set of acoustic speech stimuli and the child's range of responses, prepares relevant materials, and plans the development of the task, all according to the child's specific needs for auditory practice. 2. Attention is directed to a particular listening skill, usually represented by a single cell in the stimulus–response matrix (e.g., phrase discrimination).

Source: Erber (1982). Reprinted with permission by the Association for the Deaf.

DASL II

Stout and Windle (1994) developed a sequential, highly structured auditory-training curriculum called the Developmental Approach to Successful Listening II (DASL II). Like Erber's (1982) approach, the DASL II consists of a hierarchy of listening skills that are worked on in relatively brief, individualized sessions. The DASL II curriculum can be used with persons of any age, but it has been utilized mainly with preschool- and school-aged youngsters who use either hearing aids or cochlear implants. Three specific areas of auditory skill development are focused on the following:

- Sound awareness: Deals with the development of the basic skills of listening for both environmental and speech sounds. The care/use of hearing aids and cochlear implants is also included.
- Phonetic listening: Includes exposure to fundamental aspects of speech perception, such as the duration, intensity, pitch, and rate of speech. The discrimination and identification of vowels and consonants in isolation and in words are included in this area.
- Auditory comprehension: Emphasizes the understanding of spoken language by the child with hearing loss. Includes a wide range of auditory-processing activities, from basic discrimination of common words to comprehension of complex verbal messages in unstructured situations.

The authors developed a placement test that enables the clinician to evaluate the child's auditory skills relative to each of these three main areas. Specific subskills are tested, making it possible to determine the particular listening skills that a child has or has not acquired. As with the GASP approach, information from the DASL II placement test enables the clinician

to determine the appropriate placement of the child within the auditory skills curriculum. The test's developers provided numerous activity suggestions for the clinician. These address each of the many subskills of the three main areas of listening that make up the DASL II. These are organized from the simplest to the most difficult listening task.

The following example is a list of subskills related to sound awareness that are included in the DASL II. Similar subskill lists and related activities are provided by the developers for all components of the DASL II.

Developing Sound Awareness Subskills

1. Responds to the presence of a loud, low-frequency gross environmental sound. (Example: loud banging on a hard surface)
2. Responds to the presence of a loud speech syllable or word.
3. Responds to the presence of a variety of different gross environmental sounds.
4. Indicates when ongoing environmental sounds stop.
5. Indicates when a sustained speech syllable or word stops.
6. Indicates when teacher or parent turns both hearing aids (or processor) on or off.
7. Discriminates between presence of spoken syllable or word and silence.
8. Discriminates between a variety of familiar environmental sounds in a set of two choices.
9. Discriminates between a variety of familiar environmental sounds in a set of three choices.
10. Discriminates between a variety of environmental sounds in a set of four choices.
11. If the student is amplified binaurally, locates the direction of sound on the same plane.
12. If the student is amplified binaurally, locates the direction of sound on different planes.
13. Identifies common environmental sounds.
14. If the student is amplified binaurally, they can detect when one aid is on versus when both aids are on in a structured situation.

A team approach is encouraged with the DASL II, with the audiologist, speech-language pathologist, classroom teacher, and parents working in a coordinated fashion on relevant subskills. This makes it vital that frequent communication occurs among the team members.

SKI-HI

Home intervention involves guiding the parents as they carry out important components of an early audiologic rehabilitation program in the home for an infant diagnosed with a hearing loss.

Clark and Watkins (1985) developed this comprehensive identification and home intervention treatment curriculum for infants with hearing loss and their families, and it is widely used nationally (see Chapter 9 for more details). One of the major components of SKI-HI's treatment plan is a developmentally based auditory stimulation training program. It is utilized in conjunction with speech-language stimulation and consists of four phases and 11 general skills, which are listed in Table 4–10. Although these phases and skills are organized developmentally, infants may not always move sequentially from one phase or skill on the list to the next-highest one in a completely predictable manner. SKI-HI provides an extensive description of activities that the clinician and parent or caregiver may utilize in working on subskills related to each of the specific general skills included in each phase of the auditory training

TABLE 4–10. The Four Phases and 11 Skills of the SKI-HI Auditory Program

The approximate timeline indicates the estimated amount of time spent by a profoundly deaf infant in each phase. The age of the child upon entry into the program and the amount of hearing loss are among the factors that will affect the time needed to progress through the four phases.

Phases	Skills
Phase I (4–7 months)	1. ***Attending:*** Child is aware of presence of home and/or speech sounds but may not know meanings; stops, listens, etc. 2. ***Early vocalizing:*** Child coos, gurgles, repeats syllables, etc.
Phase II (5–16 months)	3. ***Recognizing:*** Child knows meaning of home and/or speech sounds but may not be able to locate; smiles when hears Daddy home, etc. 4. ***Locating:*** Child turns to, points to, locates sound sources 5. ***Vocalizing with inflection:*** High/low, loud/soft, up/down
Phase III (9–14 months)	6. ***Hearing at distances and levels:*** Child locates sounds far away and/or above and below 7. ***Producing some vowels and consonants***
Phase IV (12–18 months)	8. ***Environmental discrimination and comprehension:*** Child hears differences among and/or understands home sounds 9. ***Vocal discrimination and comprehension:*** Child hears differences (a) among vocal sounds, (b) among words, or (c) among phrases and/or understands them 10. ***Speech sound discrimination and comprehension:*** Child hears differences among and/or understands distinct speech sounds 11. ***Speech use:*** Child imitates and/or uses speech meaningfully

Source: Adapted from Watkins (2004) and Watkins and Clark (1993).

program. The structure and completeness of SKI-HI's auditory training component make it user friendly for parents under the guidance of clinicians. Table 4–11 provides a summary of an example of listening activities that are part of SKI-HI's comprehensive auditory stimulation program.

SPICE

Moog et al. (1995) developed the Speech Perception Instructional Curriculum and Evaluation (SPICE) to provide a guide for clinicians in evaluating and developing auditory skills in children with severe to profound hearing loss. It contains goals and objectives associated with four levels of speech perception. The first level, detection, is intended to establish an awareness and responsiveness to speech. The second and third levels, suprasegmental and vowel and consonant perception, are worked on in tandem. In the suprasegmental section, children work on differentiating speech based on gross variations in duration, stress, and intonation. In the vowel and consonant section, children begin to make perceptual distinctions among individual word stimuli with similar duration, stress, and intonation features but with different vowels and consonants. With progress, the child is introduced to the fourth level, connected speech. Now the emphasis is the perception of words in a more natural environment (phrases and sentences). Activities for SPICE are done with combined auditory–visual presentation as well as auditory-only listening situations. Many of these activities are carried out in short, structured therapy sessions that concentrate on specific listening skills. As the

Connected, or running, speech is natural or conversational speech.

TABLE 4–11. A Lesson in SKI-HI's Auditory Stimulation and Training Program

Recognition of Objects and Events From Sound Source (Phase II, Skill 3, Subskill 6)	
Parent objective	Parent will provide repeated meaningful opportunities for their child to associate environmental and speech sounds with their source.
Child objective	Child will demonstrate recognition of environmental and speech sounds by realizing their source.
Lesson	Review with the parent the sounds and activities that you have been utilizing for previous work on attending. Continue these activities, ensuring that the child is aware of the source of the sound and that the sounds are relevant to the child.
Materials	Naturally occurring environmental sounds and voice
Activities	1. Ask everyone who comes to visit to knock several times, pause, and knock again. When someone knocks, take your child to the door and say, "listen," etc. 2. Encourage the child to discover different sounds that toys make by providing playtime with several different sound toys. 3. Stimulate the child to produce sounds by manipulating objects or toys (banging pans, squeezing toy, etc.) and stimulate vocalization by making sounds as you play with the toys. 4. Imitate the child's actions, such as shaking a rattle, and imitate all vocalizations. 5. Associate speech with all major movements (e.g., saying "roll" each time you roll the child over and "up" when you pick them up). 6. Stimulate association of particular voices with particular people by having siblings/relatives use voice as they play with the child.

Source: Adapted from Clark and Watkins (1985), Watkins (2004), and Watkins and Clark (1993).

child progresses, the newly acquired skills can be refined further in more natural, informal conversation. SPICE has been used extensively with children using cochlear implants as an approach to developing listening skills in conjunction with their expanded auditory input.

Cochlear Implant Manufacturers

The major cochlear implant manufacturers have made available supportive programs for developing and improving auditory skills and facilitating the use of cochlear implants for both children and adults. These materials cover a variety of important topics, including detailed information and activities related to auditory skill development.

- Advanced Bionics:
  - http://www.hearingsuccess.com
  - http://www.thelisteningroom.com
- Cochlear Americas: http://www.cochlear.com/wps/wcm/connect/us/communication-corner
- MED-EL: https://blog.medel.com/

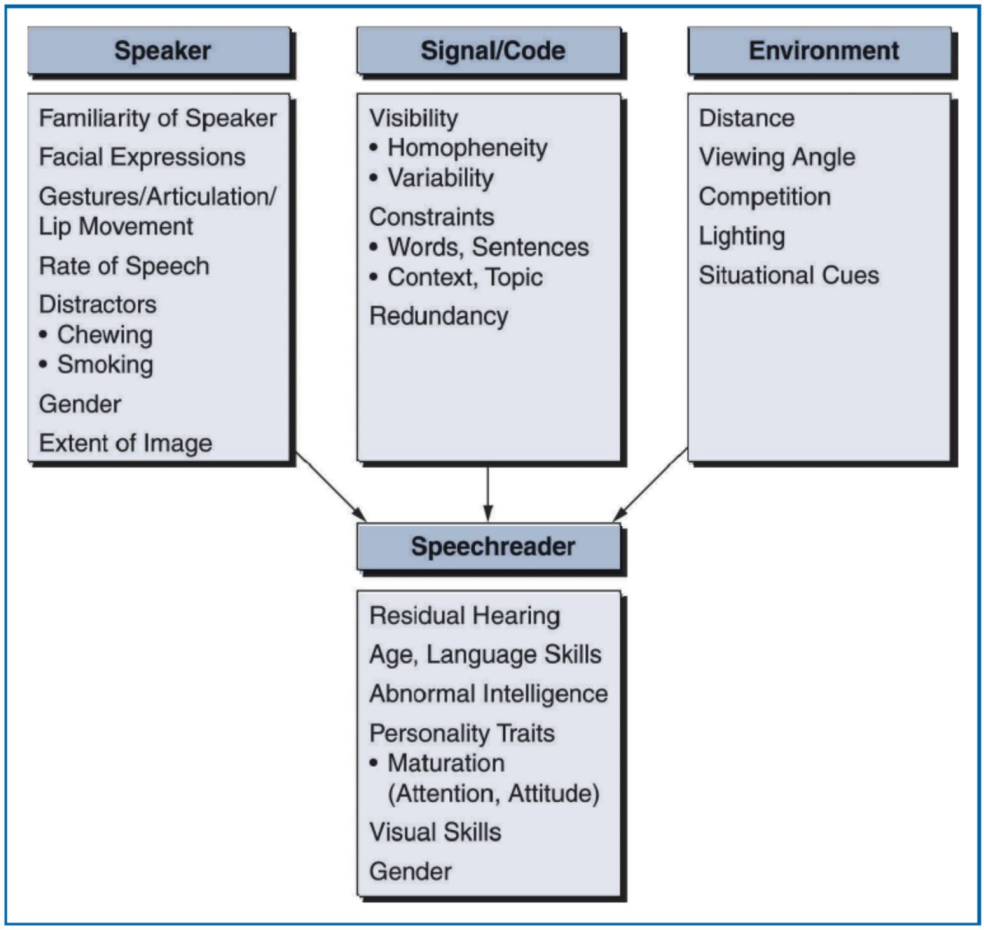


FIGURE 5-1. Summary of various factors related to speechreading performance. Arrows have been drawn from the Speaker, Signal/Code, and Environmental lists to Speechreader to signify that all these factors influence the speechreader's performance in addition to those variables that are directly related to the speechreader.

visual reception capabilities. Although the normal rate of speech may be too fast for optimal visual processing, extremely slowed and exaggerated speech production does not ensure improved comprehension. Mechanically produced slowed-down speech has actually been found to reduce speechreading abilities (IJsseldijk, 1992). It has been reported that speakers who use a slightly slower to normal speech rate accompanied by precise, not exaggerated, articulation are the easiest for the speechreader to understand. In addition, the speaker should avoid simultaneous oral activities, such as chewing, smoking, and yawning, when conversing with a person with hearing loss. Other potentially counterproductive postures include holding a hand near the mouth while talking and wearing sunglasses. While the masking effects of these coincidental activities have not been documented, they seem likely to complicate an already trying task. With respect to the gender of speakers, Daly et al. (1996) demonstrated that female speakers are easier to speechread than are male speakers. Related to this finding, little evidence exists concerning the potential influence that gender-related variables associated with the speaker, such as a mustache or the use of lipstick, may have on speechreading success.

A slightly slower to normal speech rate with precise, not exaggerated, articulation is best for speechreading. This is sometimes referred to as CLEAR speech (see Chapters 4 and 10).

TABLE 5–6. Forms of Manual and Spoken Communication

American Sign Language (ASL)	Pidgin Signed English (PSE)	Signed English	Linguistics of Visual English (LOVE)	Signing Exact English (SEE II)	Seeing Essential English (SEE I)	Finger-Spelling	Cued Speech	English
Independent language; visual manual mode; own grammar; own syntax; signs are meaning based; has dialects, regionalisms, slang, puns; can be written; wide range of vocabulary covering minute differences in meaning; may borrow from other languages; is verbal, but also makes use of nonverbal elements.	A combination of ASL and the sign systems elements, ranging from the more ASL-like (occasionally called Ameslish) to the more English-like (sometimes called CASE—Conceptually Accurate Signed English). Usually contains few if any sign markers (see Signed English), yet makes frequent use of fingerspelled English words. Used in conjunction with speech in interpreting and college teaching. Signs are meaning based.	Signed in accordance with English grammar, but signs are meaning based; specially invented sign markers for important affixes in English; invented by Bornstein; used widely in education.	Essentially the same as SEE II, but has a method of writing each sign; used in education; invented by Wampler; usage is diminishing.	Signs are word based; special signs for all affixes in English; signed in strict accordance with English; invented by Zawolkow, Pfetzing, and Gustason; widely used in education; very influential.	Signs are based on word roots (morphemes) (trans/port/a/tion); an extreme form of word-based signs; invented by Anthony; not popular in United States, but still common in Iowa and Colorado schools for the deaf; signs for all affixes.	Manual representation of the written language; one hand shape for each letter of alphabet; used to borrow English words in ASL; when used with speech and speechreading, it is called the Rochester Method.	Employs eight hand shapes in four positions on the face; used in conjunction with lip movements to enable a deaf person to lipread more easily; based on sound with the syllable as the basic unit; devised by Orin Cornett at Gallaudet College.	Independent language; aural–oral mode; own grammar; own syntax; words are meaning based; contains dialects, regionalisms, slang, puns; can be written; wide range of vocabulary covering minute differences in meaning; may borrow from other languages; is verbal but also makes use of nonverbal elements.
			Artificial pedagogical systems, invented for educational purposes					
			Nonverbal communication: natural gestures, facial expression, body movements, body language, pantomime					

Source: Based on H. Smith, personal communication, 1984.

historical and more detailed discussion of these systems. Most documentation indicates the LOVE system is no longer used.

Bilingual–Bicultural

Bilingual–bicultural (bi–bi) is a concept that an individual has some degree of proficiency in two languages as well as two cultures (Ragavan & Cowden, 2020). This method is based on the premise that children will learn ASL as the first and primary language, with English or another language taught as a second language in written form. Awareness of the concept was raised within the Deaf communities in the 1980s (Gibson et al., 1997). Valdes et al. (2015) referred to bi–bi as "multilingual deaf education." According to Kurz et al. (2021), the purpose of this method of communication is to provide an opportunity for students to be academically and linguistically competent in both ASL and English or another language. By having competence in both ASL and another language, individuals will develop connections with their family's language as well as the culture and the larger hearing world, and will form cultural, social, and linguistic relationships with other children and adults who are deaf (Kurz et al., 2021).

Fingerspelling

Fingerspelling uses 26 different hand shapes and two hand movements to represent the 26 letters of the English alphabet to allow the signer to spell words with their fingers (Moser et al., 1960). Collectively, these are also referred to as the "manual alphabet." Figure 5–7 displays the hand shapes as they appear to the communication partner. As the signs only represent letters, certain linguistic elements, like intonation, are not represented. However, it has been reported that fingerspelling is not just a manual code for spoken language, but also a naturally acquired, integrated part of the sign lexicon (Padden & Le Master, 1985; Sehyr et al., 2018). Fingerspelling is used within MCE systems as well as ASL to express proper names, technical terms, and events that cannot be conveyed by signs. Although fingerspelling is an exact and effective means of communication, it is the least efficient form of manual communication; each letter of each word must be produced, making it a relatively laborious means of communicating.

Rochester Method

The Rochester Method is the closest type of MCE to written English, using fingerspelling for every word. This system was developed in 1878 by Zenas Westervelt in the school now known as Rochester School for the Deaf (https://www.rsdeaf.org/) (McLaughlin, 1920). Use of the Rochester Method as a stand-alone MCE system has been almost discontinued due to the tedious and time-consuming process of spelling all words manually.

Cued Speech

Orin Cornett (1967) created the Cued Speech communication system to help individuals who are deaf or hard of hearing access phonology within the English language by representing the phonemes of spoken English. His vision was to facilitate rapid learning and effective communication while maintaining the natural rhythm of communication (Trezek, 2017). The Cued Speech system has three components: the mouth movements associated with speaking the language, eight handshapes or cues to represent the consonants, and four placements near the mouth to represent vowels (Shull & Crain, 2010). The National Cued Speech Association